


# USING WIRESHARK TO VIEW NETWORK TRAFFIC

## ABSTRACT

This lab details a practical Wireshark exercise capturing and analyzing ICMP data from local and remote network targets. By gathering device configurations and applying traffic filters, the exercise examines how data packets behave during communication. It contrasts local ping results against external domain requests (such as Yahoo, Cisco, and Google). The findings demonstrate how local devices use Address Resolution Protocol (ARP) to identify hardware addresses on a LAN, while external requests are directed through the default gateway.

**Allan Ngugi**
**Network Security**

**Lab: Using Wireshark to View Network Traffic**

# Table of Contents

**Introduction** ..... 3

**Answers to Questions** ..... 3

**Part 1: Capturing and Analyzing Local ICMP Data in Wireshark** ..... 3

**Step 1: Retrieve your PC interface addresses** ..... 3

**Step 2: Start Wireshark and begin capturing data** ..... 4

**Step 3: Examine the captured data** ..... 9

**Part 2: Capture and Analyze Remote ICMP Data in Wireshark** ..... 12

**Step 1: Start capturing data on the interface** ..... 12

**Step 2: Examining and analyzing the data from the remote hosts** ..... 14

**Reflection Question** ..... 15

**Appendix A: Allowing ICMP Traffic Through a Firewall** ..... 15

**Conclusion** ..... 16

**Lab: Using Wireshark to View Network Traffic**

# Introduction

This lab details a practical exercise using Wireshark to capture and analyze ICMP data generated by pinging local and remote network targets. The exercise begins by gathering local device details—including the IP address, MAC address, and default gateway—using the command prompt. It then records the process of filtering specific traffic within Wireshark to observe how data packets behave during network communication.

The analysis compares local ping results against requests sent to external domain names such as Yahoo, Cisco, and Google. By examining captured frames and checking local ARP tables, the exercise demonstrates how local devices identify hardware addresses on the LAN and how requests to external destinations are directed to the default gateway.

# Answers to Questions

## Part 1: Capturing and Analyzing Local ICMP Data in Wireshark

In Part 1 of this lab, you will ping another PC on the LAN and capture ICMP requests and replies in Wireshark. You will also look inside the frames captured for specific information. This analysis should help to clarify how packet headers are used to transport data to their destination.

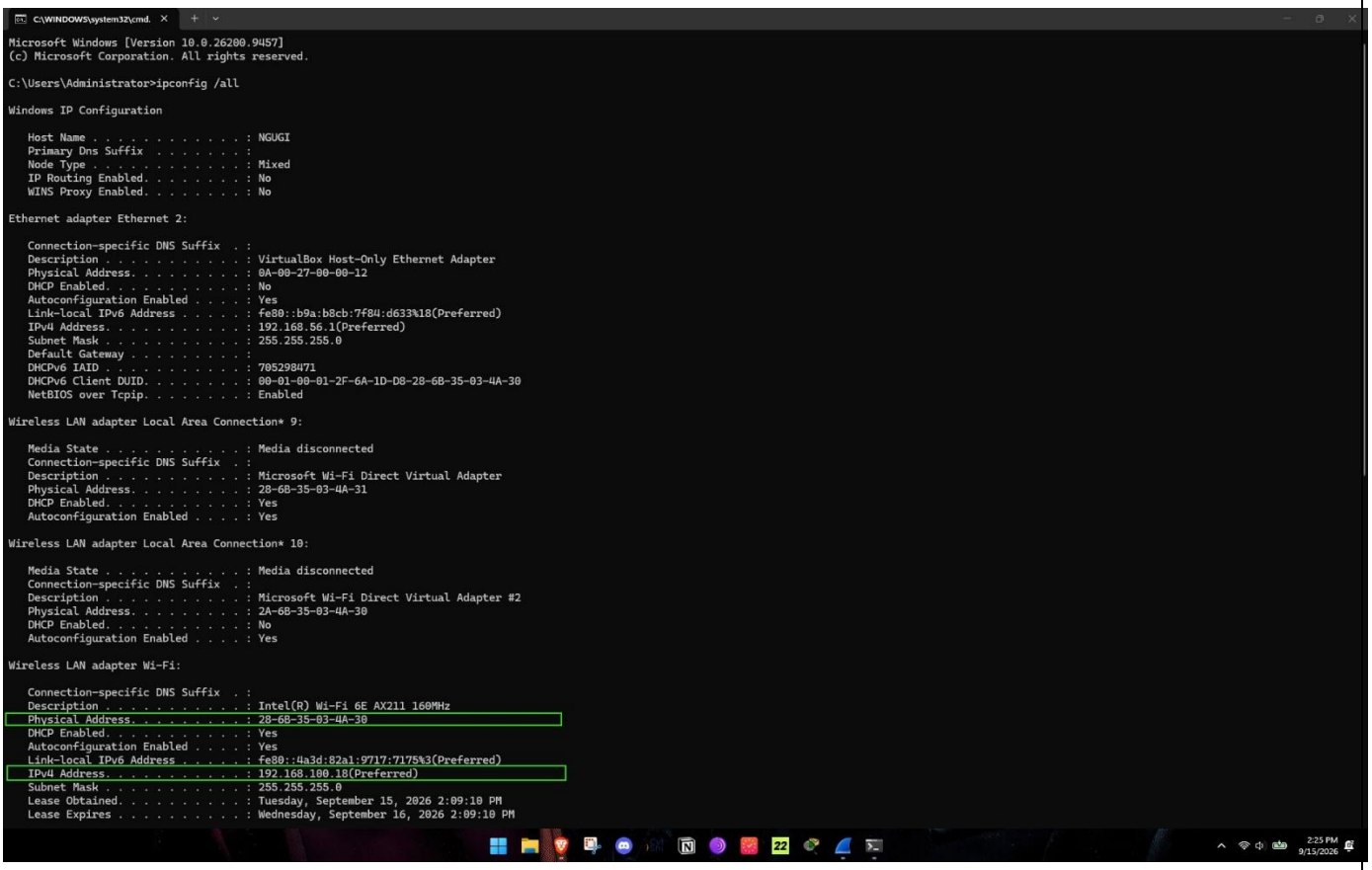
### Step 1: Retrieve your PC interface addresses

For this lab, you will need to retrieve your PC IP address and its network interface card (NIC) physical address, also called the MAC address.

a. Open a command window, type **ipconfig /all**, and then press Enter.

b. Note the IP address of your PC interface, its description, and its MAC (physical) address.

# Lab: Using Wireshark to View Network Traffic



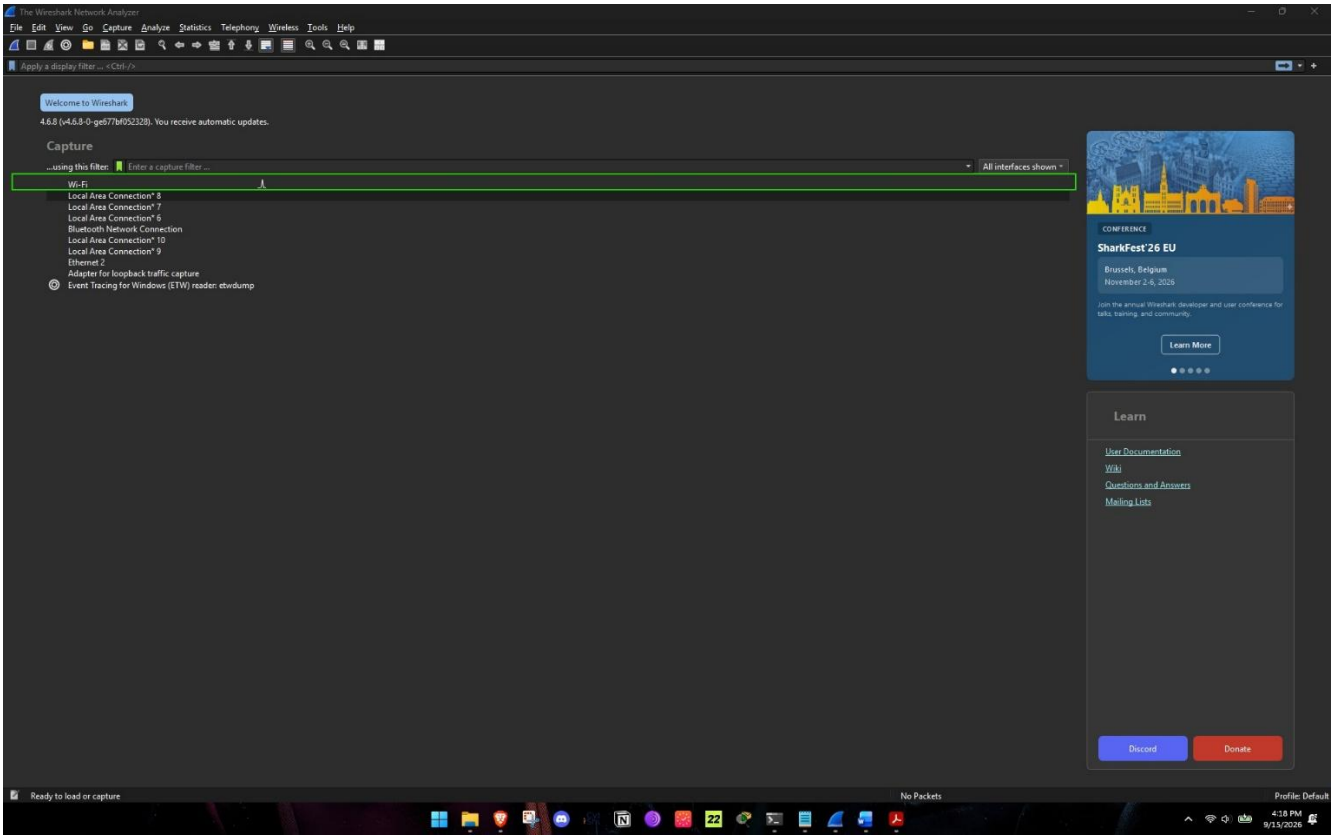

c. Ask a team member or team members for their PC IP address and provide your PC IP address to them. Do not provide them with your MAC address at this time

(This was quite impossible on my end due to the fact that I did this solely, without having a partner whom we could exchange IP addresses with. I therefore went with the option of utilizing my Wi-Fi router).

## Step 2: Start Wireshark and begin capturing data

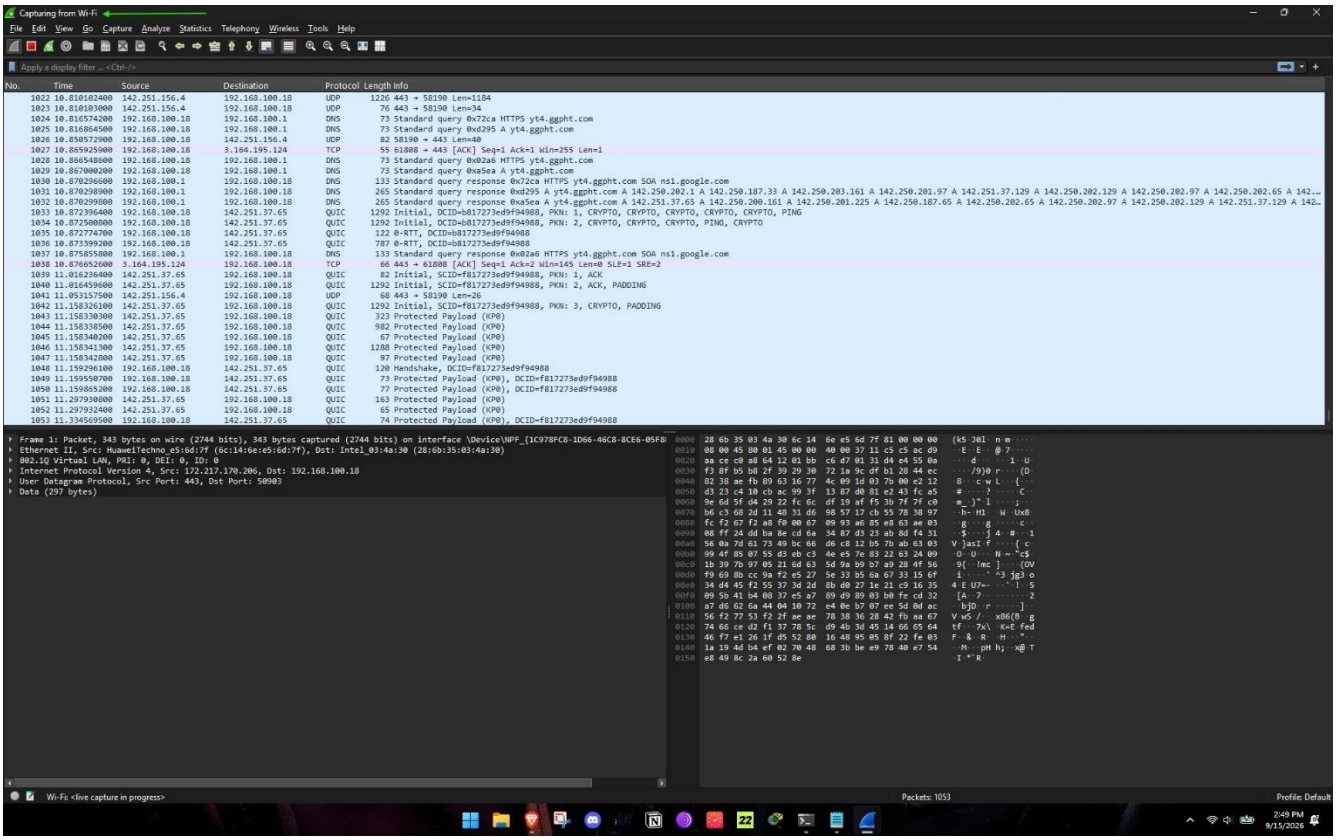
a. Navigate to Wireshark. Double-click the desired interface to start the packet capture. Make sure the desired interface has traffic.

# Lab: Using Wireshark to View Network Traffic




*Wi-Fi was the desired capture interface, and traffic is shown below:*

# Lab: Using Wireshark to View Network Traffic

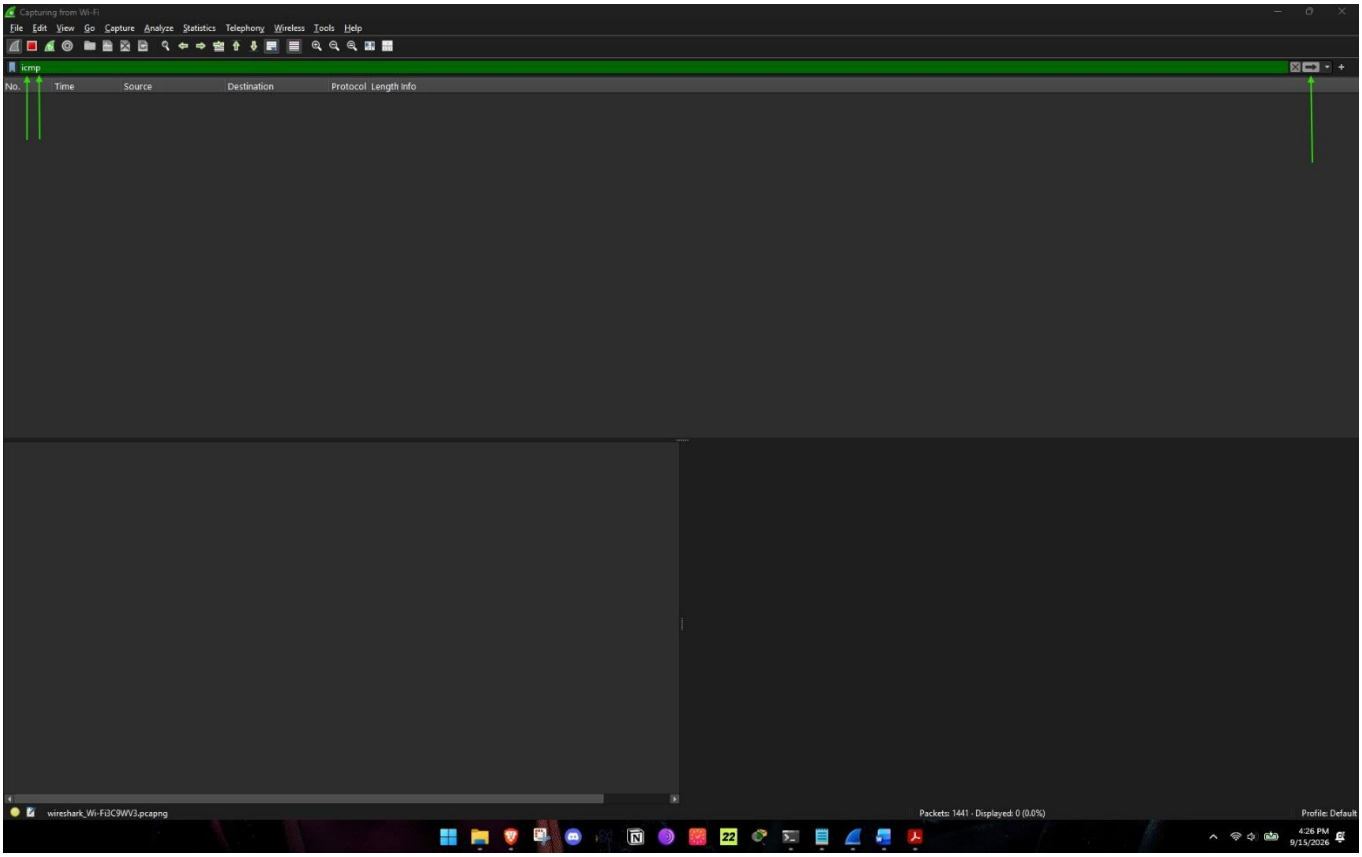



b. Information will start scrolling down the top section in Wireshark. The data lines will appear in different colours based on protocol.

   This information can scroll by very quickly depending on what communication is taking place between the PC and the router. We can apply a filter to make it easier to view and work with the data that is being captured by Wireshark.

   For this lab, we are only interested in displaying ICMP (ping) PDUs. Type icmp in the filter box at the top of Wireshark and press Enter, or click the Apply button (arrow sign) to view only ICP (ping) PDUs.

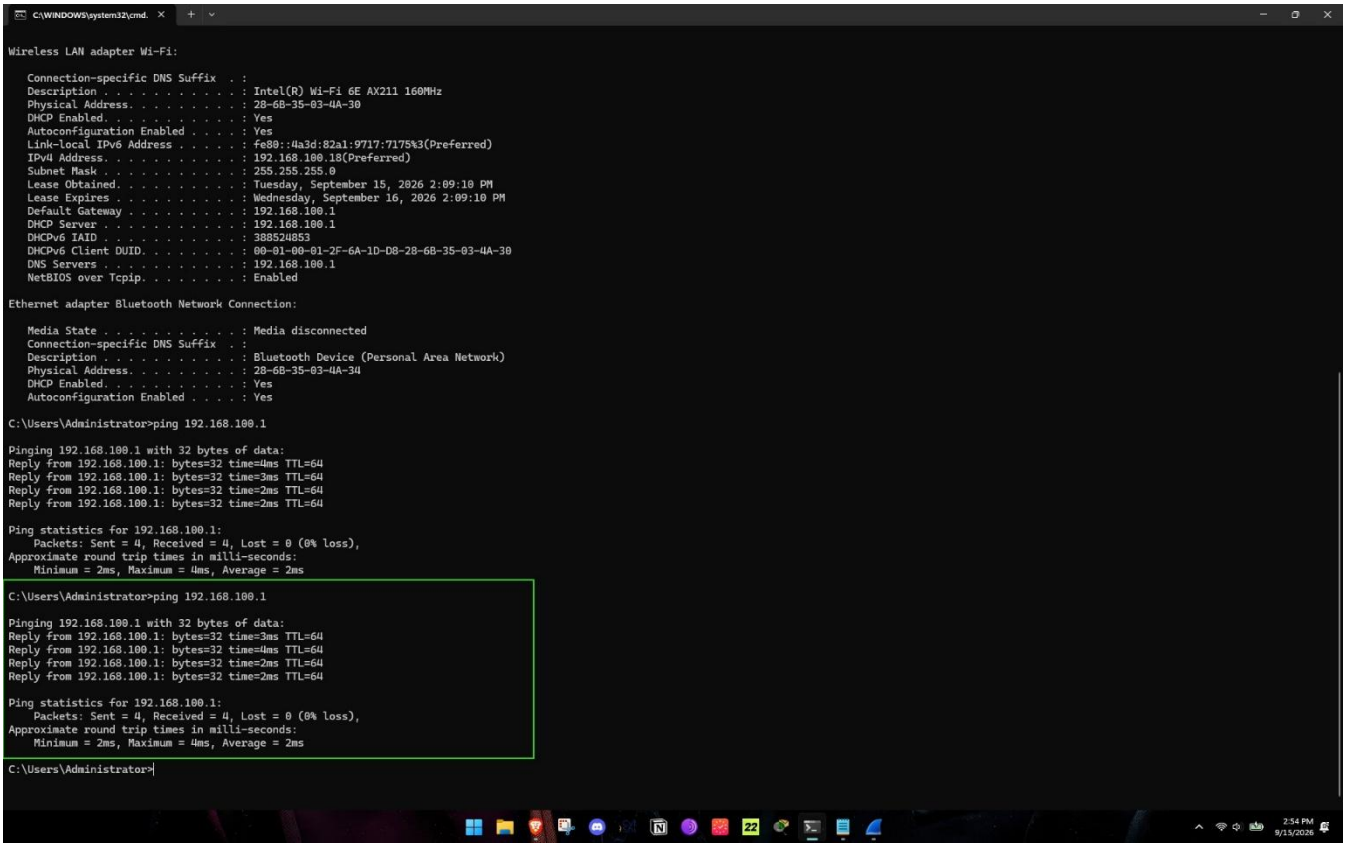
# Lab: Using Wireshark to View Network Traffic




*The filter icmp is applied in Wireshark, while packets are still being captured in the background.*

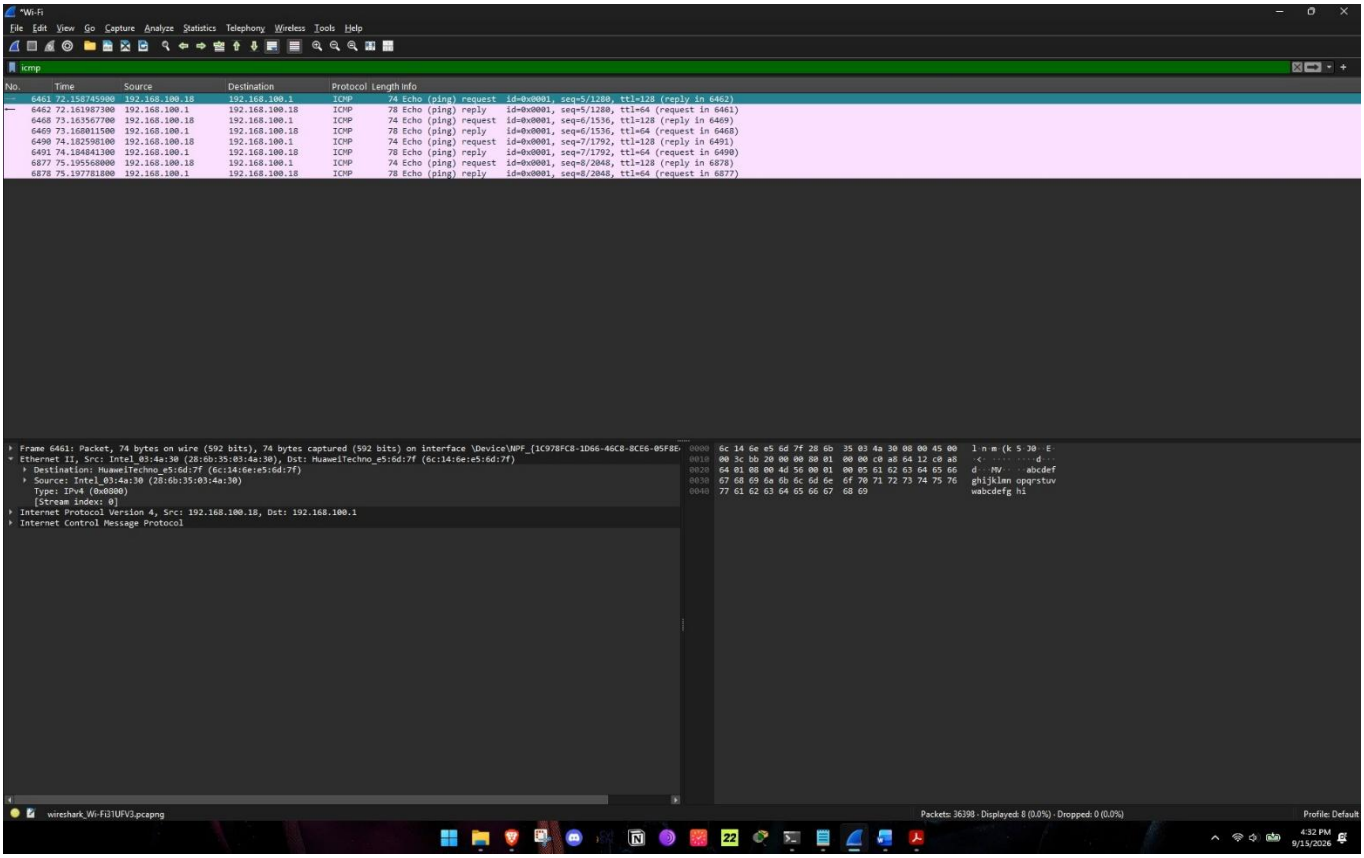
c. This filter causes all data in the top window to disappear, but you are still capturing the traffic on the interface. Navigate to a command prompt window and ping the default gateway.

# Lab: Using Wireshark to View Network Traffic




Notice that you start seeing data appear in the top window of Wireshark again.

# Lab: Using Wireshark to View Network Traffic




d. Stop capturing data by clicking the Stop Capture icon.

## Step 3: Examine the captured data

In Step 3, examine the data that was generated by the ping requests of the gateway. Wireshark data is displayed in three sections: 1) The top section displays the list of PDU frames captured with a summary of the IP packet information listed; 2) the middle section lists PDU information for the frame selected in the top part of the screen and separates a captured PDU frame by its protocol layers; and 3) the bottom section displays the raw data of each layer. The raw data is displayed in both hexadecimal and decimal form.

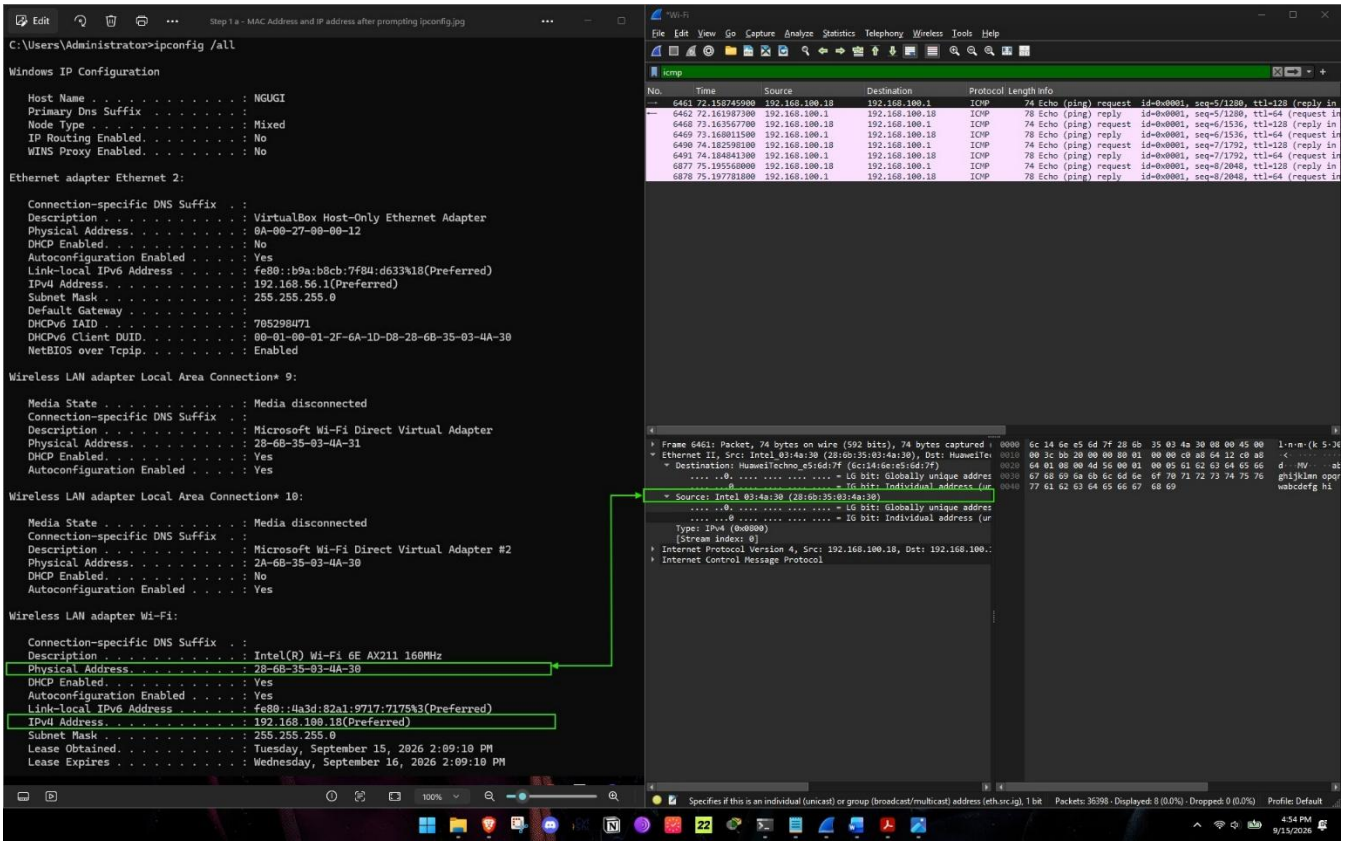
a. Click the first ICMP request PDU frames in the top section of Wireshark. Notice that the **Source** column has your PC IP address, and the **Destination** column contains the IP address of the router (default gateway).

# Lab: Using Wireshark to View Network Traffic

b. With this PDU frame still selected in the top section, navigate to the middle section. Click the plus sign to the left of the Ethernet II row to view the destination and source MAC addresses.

**Does the source MAC address match your PC interface?**

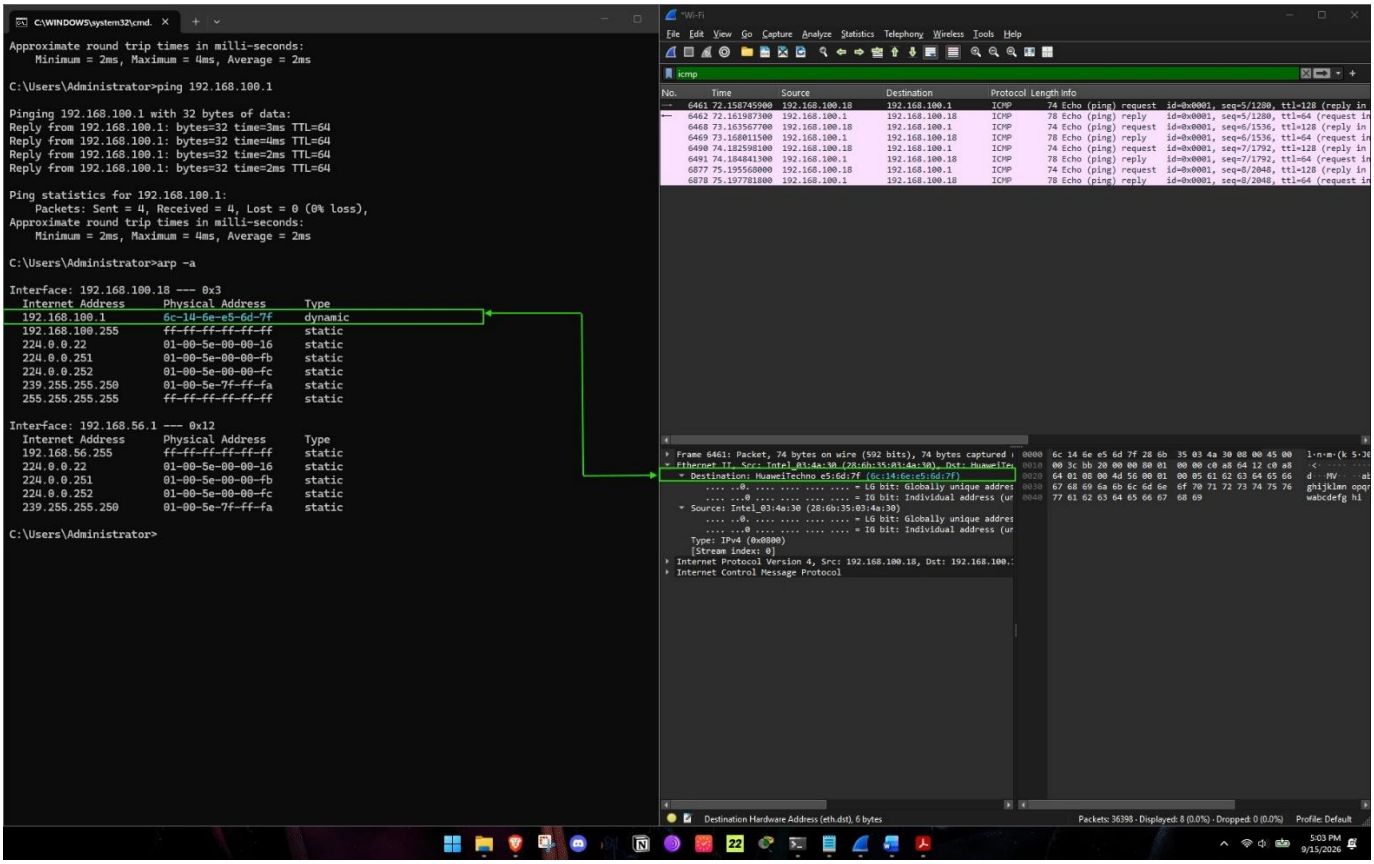
*Yes, the source MAC address matches my PC interface. This is shown below:*




**Does the destination MAC address in Wireshark match the MAC address of the router?**

*Yes, in Wireshark, the MAC address is the destination address. I was able to confirm by going to the command prompt, and typing arp -a to generate the ARP (Address Resolution Protocol) cache table, which lists the mapping of IP addresses to MAC addresses in my local network.*

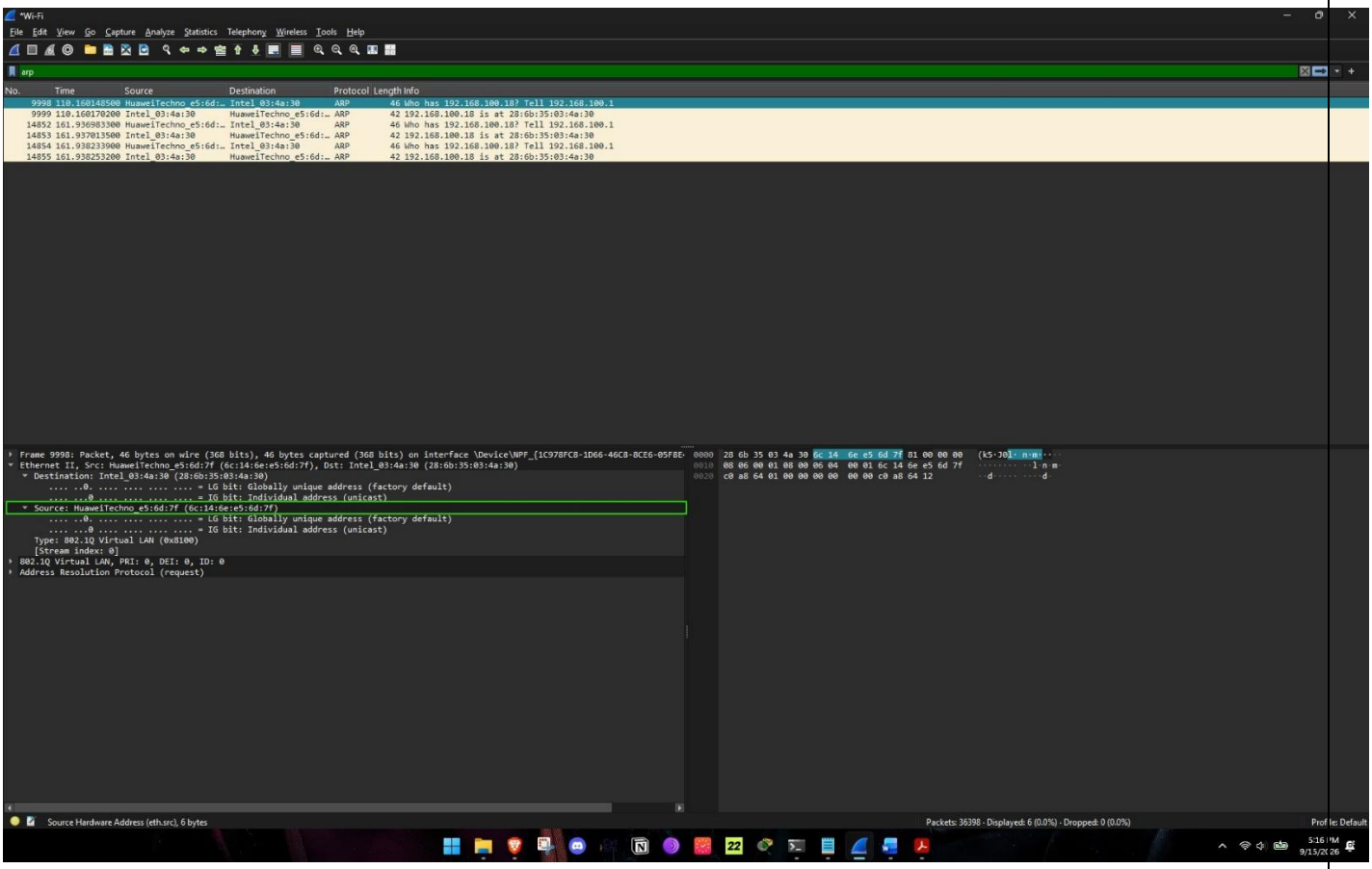
# Lab: Using Wireshark to View Network Traffic




**How is the MAC address of the pinged router obtained by your PC?**

*The MAC address of the pinged router is obtained by my PC through an ARP request. This can be proven by filtering arp in Wireshark as shown below:*

## Lab: Using Wireshark to View Network Traffic




**Note:** In the preceding example of a captured ICMP request, ICMP data is encapsulated inside an IPv4 packet PDU (IPv4 header) which is then encapsulated in an Ethernet II frame PDU (Ethernet II header) for transmission on the LAN.

# Part 2: Capture and Analyze Remote ICMP Data in Wireshark

In Part 2, you will ping remote hosts (hosts not on the LAN) and examine the generated data from those pings. You will then determine what is different about this data from the data examined in Part 1.

## Step 1: Start capturing data on the interface

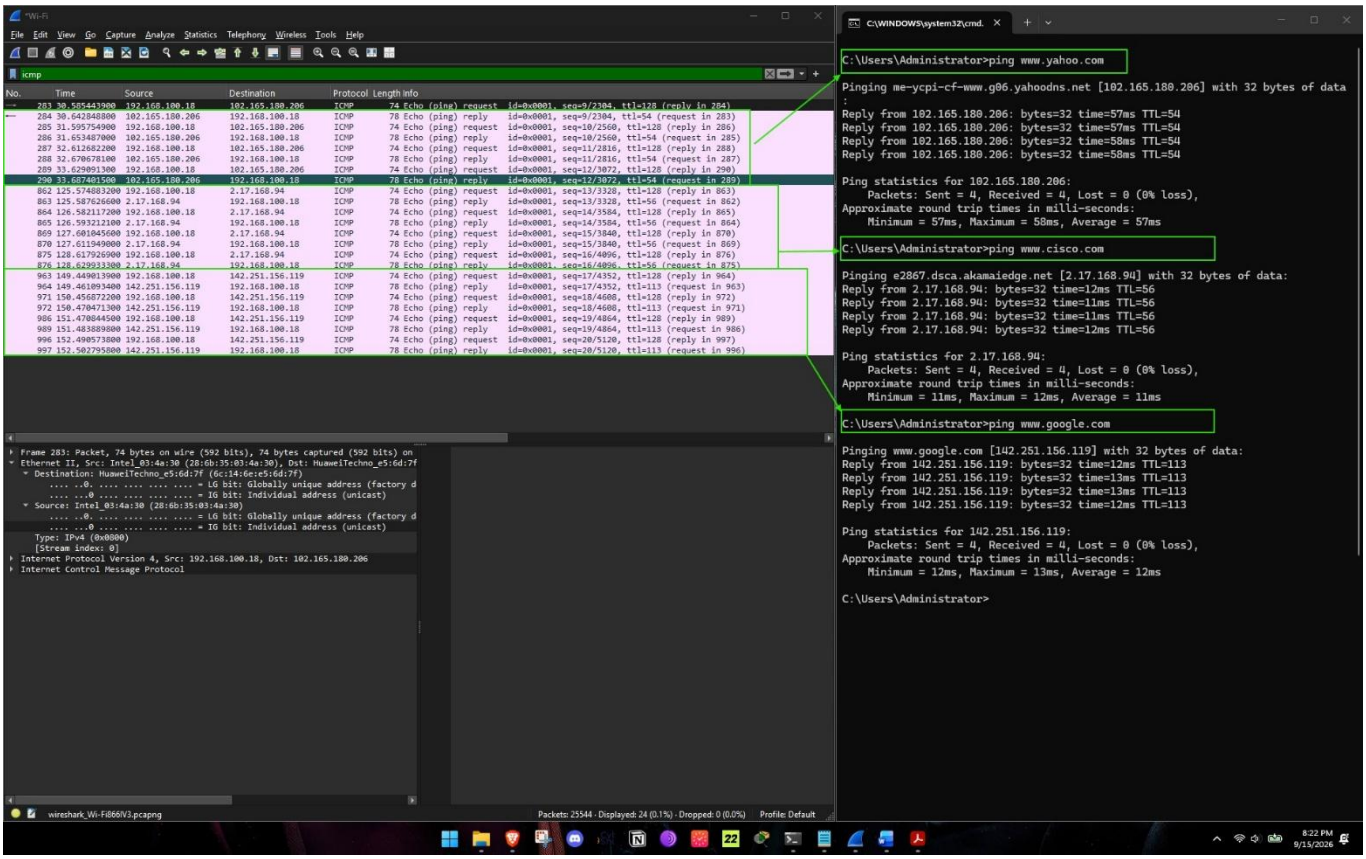
a. Start the data capture again.

# Lab: Using Wireshark to View Network Traffic

b. A window prompts you to save the previously captured data before starting another capture. It is not necessary to save this data. Cluck continue without saving.

c. With the capture active, ping the following three website URLs:

1. www.yahoo.com
2. www.cisco.com
3. www.google.com




**Note**: When you ping the URLs listed, notice that the Domain Name Server (DNS) translates the URL to an IP address. Note the IP address received for each URL.

d. You can stop capturing data by clicking the Stop Capture icon.

# Lab: Using Wireshark to View Network Traffic

## Step 2: Examining and Analyzing the data from the remote hosts

a. Review the captured data in Wireshark and examine the IP and MAC addresses of the three locations that you pinged. List the destination IP and MAC addresses for all three locations in the space provided.

**IP address for www.yahoo.com:** *102.165.180.206*

**MAC address for www.yahoo.com:** *6c-14-6e-e5-6d-7f* (default gateway)

**IP address for www.cisco.com:** *2.17.168.94*

**MAC address for www.cisco.com:** *6c-14-6e-e5-6d-7f* (default gateway)

**IP address for www.google.com:** *142.251.156.119*

**MAC address for www.google.com:** *6c-14-6e-e5-6d-7f* (default gateway)

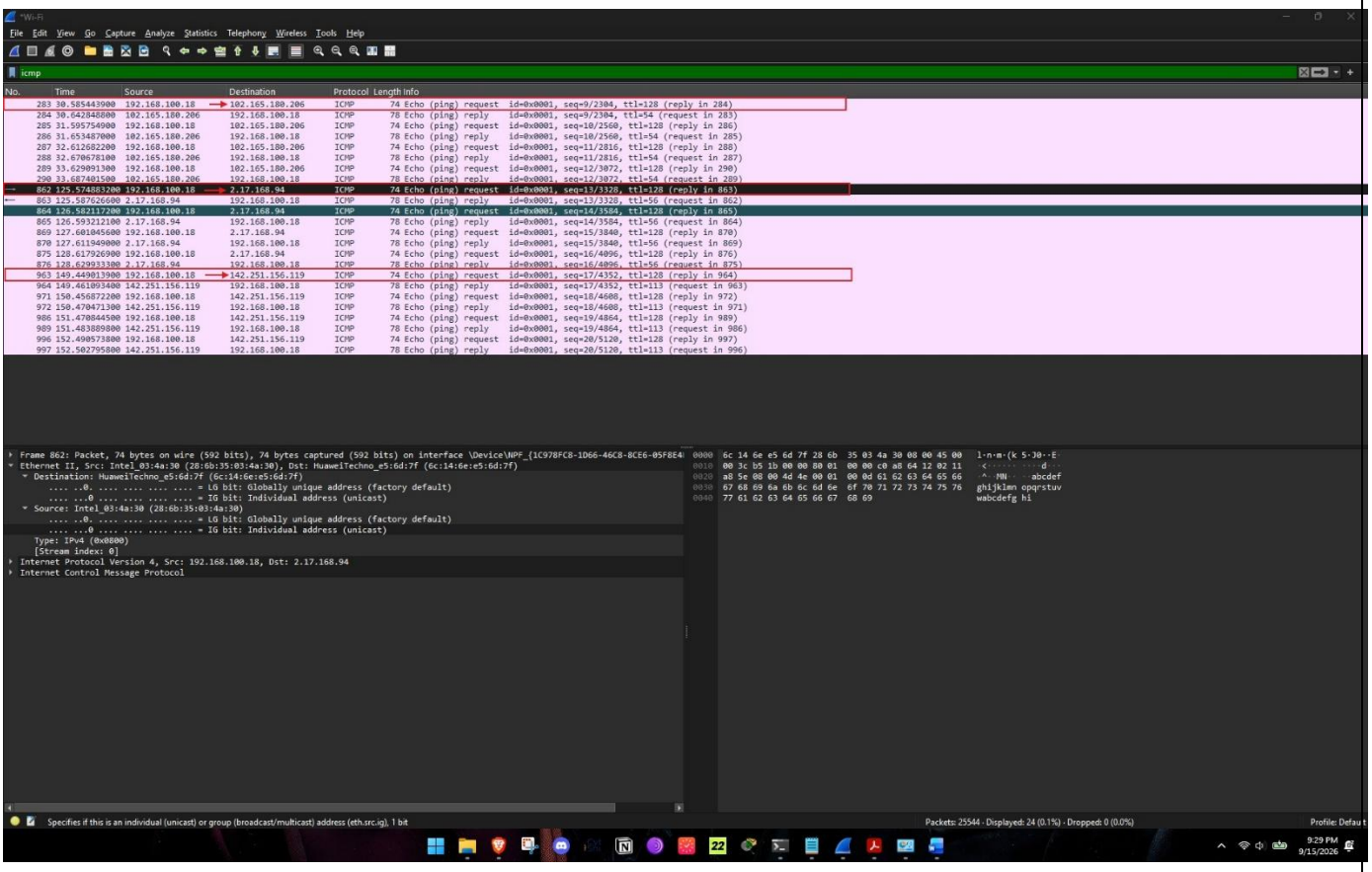



*Image showing the pinging and the results of the remote hosts*

## Lab: Using Wireshark to View Network Traffic

**b. What is significant about this information?**

*The significance is in the fact that the MAC addresses for all these remote hosts are the same. They all point towards the default gateway (which is the physical or MAC address of our router).*

**c. How does this information differ from the local ping information you received in Part 1?**

*Pinging locally usually returns MAC address of the PC's Network Interface Card (NIC). On the other hand, pinging a remote host usually returns the MAC address of your default gateway.*

### Reflection Question

**Why does Wireshark show the actual MAC address of the local hosts, but not the actual MAC address for the remote hosts?**

*Wireshark shows the actual MAC addresses of local hosts because MAC addresses are used for communication within the local network, but for remote hosts, routers replace the MAC addresses at each network hop, so your device only sees the MAC address of its local gateway rather than the remote host's actual MAC address.*

# Appendix A: Allowing ICMP Traffic Through a Firewall

Since the exercise was done solely, the appendix part was omitted, since it focused on the members of the team who shared their IP addresses with each other on the same Local Area Network.

**Lab: Using Wireshark to View Network Traffic**

# Conclusion

In conclusion, the lab shows that pinging a local target reveals the actual MAC address of the destination device, whereas pinging remote hosts returns only the MAC address of the default gateway. This is because local devices rely on ARP requests to find MAC addresses within the local network, communication sent to outside destinations is handled by the local router, which replaces MAC addresses at each network hop.